

Optical Model-Driven Sharpness Mapping for Autofocus in Small Depth-of-Field and Severe Defocus Scenarios

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Abstract

Autofocus (AF) is essential for imaging systems, particularly in industrial applications such as automated optical inspection (AOI), where achieving precise focus is critical. Conventional AF methods rely on peak-searching algorithms that require dense focal sampling, making them inefficient in small depth-of-field (DoF) scenarios. Deep learning (DL)-based AF methods, while effective in general imaging, have a limited working range in small DoF conditions due to defocus uncertainty.

In this work, we propose a novel AF framework that integrates an optical model-based sharpness indicator with a deep learning approach to predict sharpness from defocused images. We leverage sharpness estimation as a reliable focus measure and apply an adaptive adjustment algorithm to adjust the focus position based on the sharpness-to-distance mapping. This method effectively addresses defocus uncertainty and enables robust autofocus across a $35\times$ DoF range. Experimental results on an AOI system demonstrate that our approach achieves reliable autofocus even from highly defocused starting points and remains robust across different textures and illumination conditions. Compared to conventional and existing DL-based approaches, our method offers improved precision, efficiency, and adaptability, making it suitable for industrial applications and small DoF scenarios.

1. Introduction

Autofocus (AF) is essential for automatically capturing high-quality images in applications such as consumer photography, scientific imaging, and automated optical inspection (AOI). Existing AF systems rely on either hardware-assisted or image-based approaches. Hardware-assisted methods use external sensors such as ultrasonic, LiDAR, or laser-based range finders to measure object distance and adjust focus [34, 39], or employ specialized designs like

dual-pixel sensors to achieve phase-detection AF [6]. While effective, these approaches increase system complexity and cost. Image-based AF techniques have been developed to overcome these limitations, utilizing high-frequency image features such as gradients, Laplacians, and variance to estimate sharpness and identify the optimal focus position. Although adaptable and functional without training under severe defocus, image-based AF methods often use peak-searching algorithms [40] requiring dense focal sampling. This is due to the algorithm's sensitivity to texture and illumination variations, which causes inconsistent sharpness predictions across different surfaces and lighting conditions, leading to inefficiency in small depth-of-field (DoF) scenarios where fine-grained adjustments are crucial for accurate focusing.

Recent advances in deep learning (DL) have transformed autofocus by enabling image-based models that interpret focus cues without extensive sampling. Current DL-based AF methods typically rely on depth estimation (DE) models in an end-to-end framework, predicting focus distance from defocused images and adjusting focus accordingly. Most are designed for general scenes, like smartphone cameras, where severe defocus is rare [6, 10]. A few DL-based AF methods address small DoF applications, such as microscopic bio-imaging [17], but struggle with defocus uncertainty, where the same defocus level occurs at two positions along the optical axis. This uncertainty limits their working range, reducing AF reliability under severe defocus.

In this study, we propose a novel autofocus method that integrates an optical defocus model with a deep learning-based sharpness prediction network. Instead of relying on direct end-to-end distance estimation, our approach utilizes optics-based sharpness as an intermediate representation, leading to more reliable autofocus under severe defocus conditions. The proposed sharpness indicator is robust to variations in texture and helps mitigate defocus uncertainty during focus adjustment, improving accuracy across different surface characteristics. Coupled with an adaptive adjustment algorithm, our method enables efficient autofocus even from highly defocused starting positions in shallow

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DoF environments. The main contributions of this work can be summarized as follows:

- Propose an optical model-based sharpness indicator capable of representing defocus levels, even in severely defocused images, and utilize a deep learning model to predict sharpness from defocused inputs.
- Develop an adaptive autofocus adjustment strategy that leverages the optical model-based sharpness to effectively overcome defocus uncertainty and reduce the number of required focus steps across a $35\times$ DoF range.
- Conduct experiments on an AOI system, achieving over 95% in-focus accuracy within five iterations and outperforming baselines in efficiency and adaptability across various textures and illumination conditions.

2. Related Works

2.1. Image based Autofocus

Conventional autofocus (AF) methods assume that objects approaching the focal plane exhibit sharper high-frequency features. These methods calculate image sharpness based on these features, leading to notable variability across different scenes due to dependence on specific image patterns. Studies have shown that the Tenenbaum Gradient [33] and Auto-Correlation [29] were among the top-performing sharpness algorithms. Later methods, such as Tenengrad Variance [24], Laplacian Variance [24], and Modified DCT [16], continued to define sharpness through high-frequency content extraction. Peak-searching techniques are essential for maximizing sharpness across image stacks captured at various focus settings. Common approaches include rule-based methods [14], Fibonacci search [15], and coarse-to-fine search [40], the most robust among conventional AF techniques. However, all searching methods require dense sampling for optimal focus.

Recent deep learning (DL)-based AF methods reduce sampling by predicting focus distance from defocused images. Herrmann et al. estimated focus distance via classification over 50 intervals [10], while Choi et al. integrated positional encoding with a classification model for dual-pixel imaging [6]. Wang et al. used a CNN to regress distance and image features for AF [35]. DL-based AF has also been applied to small DoF settings, such as bio-imaging in microscopy [17, 25, 30, 31], but often suffers from limited working ranges. Shajkofci et al. demonstrated AF within an $11\times$ DoF range [31], while Liao et al. achieved one-shot AF within $7.2\times$ DoF [17]. In contrast, our method achieves reliable autofocus over a $35\times$ DoF range.

2.2. Distance Estimation

Distance or depth estimation (DE) predicts depth from single or multiple images and plays a crucial role in DL-based autofocus, despite not being originally designed for

this purpose. Both conventional and DL-based DE methods utilize focus and defocus cues, commonly categorized as Depth from Focus (DFF) and Depth from Defocus (DFD). Conventional DFF methods [5, 8, 21, 23] estimate depth by identifying the sharpest pixels across focus stacks but require dense sampling. In contrast, DL-based approaches achieve higher accuracy with fewer sampled focal planes [9, 20, 32], with Yang et al. further improving performance by integrating Focus Volume [37]. DFD methods typically use a single defocused image to predict depth. The U-Net backbone is widely employed for DFD tasks [11], with Carvalho et al. proposing a U-Net combined with DenseNet-121 for defocus-based depth estimation [3], a design widely adopted in later studies [19, 22, 38]. Beyond DFF and DFD, DL-based DE can integrate geometric and color cues, allowing depth estimation beyond defocus information [1, 13, 18, 41]. However, related studies on DL-based DE for industrial applications remain limited, particularly in addressing severe defocus, small DoF, and environments with minimal geometric and color information.

3. Methodology

3.1. Overview

In our method, we first propose a sharpness indicator based on the defocus optical model, and then utilizing a deep learning model predicts the sharpness from defocus images. The sharpness indicator is capable of representing the defocus level of images. Next we propose an adjustment strategy, it utilizes distance mapping results from the sharpness for position adjusting and finally achieving autofocus, the framework of the proposed method is presented in Fig. 1. The defocus model is introduced in Sec. 3.2. The mapping relation between sharpness and distance is in Sec. 3.3. And finally the adjustment algorithm is presented in Sec. 3.4.

3.2. Optical Model of Defocus

The optical model underlying our AF approach defines the relationship between sharpness and defocus using the Circle of Confusion (CoC). In-focus images are produced when objects lie within the imaging system's depth-of-field, where their light rays converge to a point on the sensor plane. If the object lies outside the DoF, these light rays form a circular blur, the CoC, on the imaging plane. The definition of diameter of CoC D_{coc} is defined [38]:

$$D_{coc} = \frac{|d_o - d_{of}| \cdot f}{d_o \cdot (d_{of} - f) \cdot F} \quad (1)$$

The CoC diameter is a function of focal length f , the object distance d_o , the focal plane of the object d_{of} and finally the aperture F . The relationship with object distance is sketched in Fig. 2. In particular, the same level of blur can occur on both sides of the focal plane, introducing uncertainty that presents challenges for both AF and DE tasks.

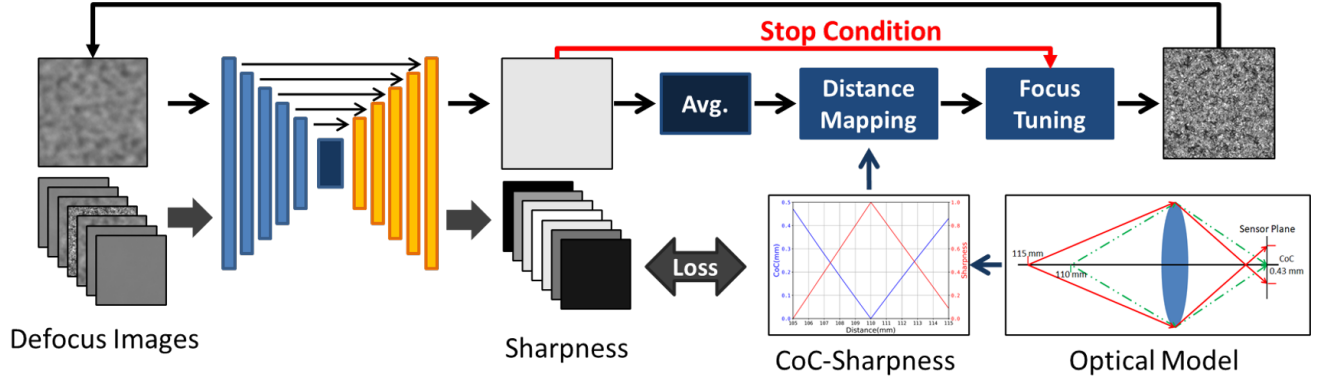


Figure 1. Framework of the proposed autofocus method, integrating an optical model-based sharpness indicator with a deep learning approach for focus adjustment.

The core of our AF method lies in predicting the relative distance d_{rel} between d_o and d_{of} , where $d_{rel} = d_{of} - d_o$. For the convenience, we introduce a constant $K = \frac{f}{(d_{of} - f) \cdot F}$, and refine the Eq. 1 to:

$$D_{coc} = \frac{d_{rel} \cdot K}{d_{of} \pm d_{rel}} \quad (2)$$

The spread of the Circle of Confusion from a point to a circular blur is the fundamental cause of image defocus. This effect can be approximated by convolving the in-focus image with a Gaussian-distributed disk mask, representing the blur introduced by CoC expansion. The mathematical representation is given by:

$$I_d = I_0 * G(D_{coc}) + noise \quad (3)$$

where I_d is the image captured at a position corresponding to d_{rel} , and I_0 is the image captured at the focal point, where $d_{rel} = 0$. $G(D_{coc})$ represents the defocus Gaussian blur kernel, determined by the diameter of the CoC, D_{coc} , and $*$ denoted as convolution. This process acts as the point-spread function (PSF), a widely used model for describing defocus phenomena. The DL-model learns to quantify sharpness from the defocus level.

The range where the CoC is below a certain threshold is called depth-of-field (DoF). The threshold is the diameter of the permissible CoC [7], which is defined as:

$$CoC_{permissible} = \frac{P_{diagonal}}{N} \quad (4)$$

where the $P_{diagonal}$ represents the diagonal length of the camera sensor plane, and N is a constant. The technical report [7] provides a data point in which $N = 1300$. Focus is achieved when the CoC remains within the permissible value. Fig. 2 illustrates the permissible CoC for lenses with identical parameters but varying working distances. The DoF expands as the working distance increases. Most imaging systems operate at working distances of over a meter, sometimes even to kilometers, which explains why general imaging systems rarely encounter severe defocus blur.

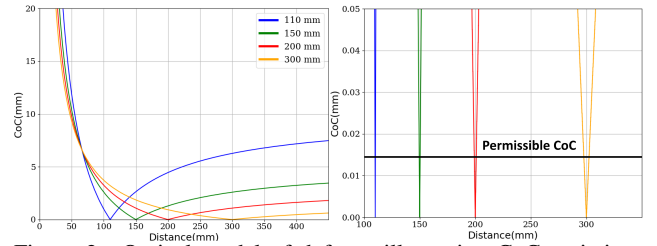


Figure 2. Optical model of defocus illustrating CoC variations across different working distances and the relationship between DoF and permissible CoC.

3.3. Sharpness based Distance Estimation

Sharpness Definition: Conventionally, sharpness is defined by high-frequency content within an image, which is highly sensitive to factors like illumination and textures. Our method uses the definition of sharpness based on the CoC model presented in Sec. 3.2. Since CoC directly correlates with the defocus level of images, we define maximum sharpness at the smallest CoC diameter and minimum sharpness at the largest CoC within the measurement range. As illustrated in Fig. 3a, the largest CoC occurs on the far left side of the range. The mathematical definition of sharpness s is given by:

$$s = S(d_{rel}) = \frac{D_{max} - D_{coc}(d_{rel})}{D_{max} - D_{min}} \quad (5)$$

where $D_{coc}(d_{rel})$ is the diameter of the CoC as a function of d_{rel} , which represents the distance to the focal plane. In the AF applications, the focal plane is assumed to be within the focus range; therefore, D_{min} is 0, corresponding to the maximum sharpness value of 1, and D_{max} corresponds to the CoC at the boundary on the near side of the lens, where the sharpness at this boundary is 0. The sharpness value can represent the defocus level of the images as Fig. 3c shows.

Sharpness Regression: We use a deep learning model to estimate sharpness as defined in Eq. 5, serving as an intermediate indicator for autofocus. Among tested architectures, U2-Net performed best for sharpness regression. The detailed experimental comparison is presented in Sec. 4.6.

For loss function, we empirically determined that a combination of L1 loss and PatchGAN loss [12] provided favorable results for sharpness estimation. This finding aligns with previous research [4], highlighting that L1 loss and GAN-based losses each have their merits in related regression tasks. Assume the predicted sharpness $\hat{s}_i = U(i_d)$ where U is the U2-Net model and i_d is the defocused image at position d_{rel} . The math representation of the loss function is listed below:

$$L_{L1} = \sum_{i=1}^N |s_d - \hat{s}_i| \quad (6)$$

$$L_{total} = L_{patchGAN} + constant \cdot L_{L1} \quad (7)$$

Where $L_{patchGAN}$ is the patchGAN loss. s_d is the predefine sharpness from Eq. 5 at distance d_{rel} . Accurate sharpness estimation is crucial not only for distance estimation but also as a stop condition for our AF framework.

Distance Mapping: Fig. 3a and Fig. 3b shows the calculated sharpness and an example of estimated sharpness from the U2-Net model of a specific range. Once the sharpness of the defocused images is determined, the distance can be directly mapped through the inverse function of sharpness. The inverse function can be derived from Eq. 2 and Eq. 5. The D_{min} in Eq. 5 is 0. We rewrite Eq. 5 to $D_{coc}(d_{rel}) = D_{max} \cdot (1 - s)$. And substitute Eq. 2 to get:

$$\frac{d_{rel} \cdot K}{d_{of} \pm d_{rel}} = D_{max} \cdot (1 - s) \quad (8)$$

We finally get the inverse function of sharpness s :

$$d_{rel} = S^{-1}(s) = \frac{D_{max} \cdot (1 - s) \cdot d_{of}}{K \pm D_{max} \cdot (1 - s)} \quad (9)$$

As shown in Sec. 3.2, the same defocus level can occur on both sides of the focal plane, causing a given sharpness value to correspond to two possible relative distances d_{rel} . In the next section, we will address direction determination during focus adjustment.

3.4. Focus Adjustment

Focus adjustment is guided by the distance mapping results described in Sec. 3.3, with adjustments made based on the estimated distance. One challenge in this AF process is to overcome the uncertainty of defocus. To resolve this, we implement an efficient solution by introducing *Take One More Step*: if the sharpness decreases in the second sampling, the adjustment moves away from the focal point; if it increases, the adjustment moves closer. Although this approach requires an extra sampling step, it significantly enhances the adaptability of our model, allowing the DL-model to focus exclusively on sharpness regression without the added burden of predicting direction.

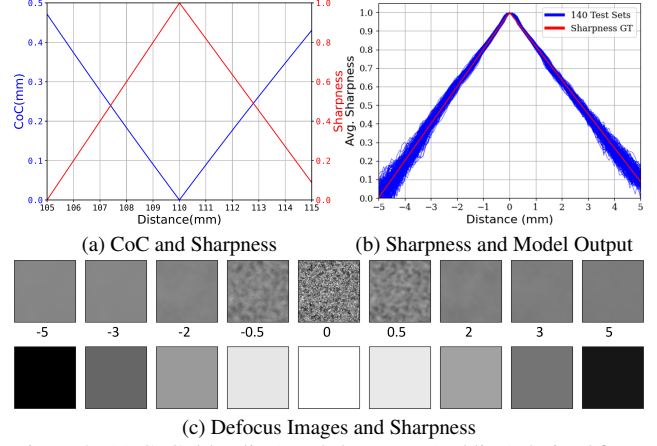


Figure 3. (a) CoC (blue line) and sharpness (red line) derived from Eq. 5. (b) Range shifted from -5 to 5 mm, where 0 mm means focal plane. (c) Anodized samples at different distances and sharpness.

Algorithm 1 Autofocus Adjustment Process

Require: Sharpness Model M , Sharpness Inverse Function S^{-1} , Image img , Max Steps N , Threshold T

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1:  $steps \leftarrow 0, prev \leftarrow 0, direction \leftarrow 1$ 
2: while True do
3:    $direction \leftarrow 1$ 
4:    $sharp \leftarrow M(Acquire(img))$ 
5:    $dist \leftarrow S^{-1}(sharp, direction)$ 
6:   if  $steps = 0$  then
7:     Adjust focus by  $dist$ 
8:   else if  $sharp < prev$  then
9:      $direction \leftarrow -1$ 
10:     $dist \leftarrow S^{-1}(prev, direction)$ 
11:    Adjust focus by  $dist$ 
12:   else
13:     Adjust focus by  $dist$ 
14:   end if
15:   if  $sharp \geq T$  or  $step \geq N$  then
16:     return
17:   end if
18:    $prev \leftarrow sharp, steps \leftarrow steps + 1$ 
19: end while

```

In an ideal scenario where distance estimation is perfect, only a single adjustment would be required. However, due to inherent model uncertainties and defocus uncertainty in the optical model, multiple tuning steps are typically needed until the sharpness reaches a defined threshold. The pseudo-code for this process is provided in Alg. 1.

The $\pm$ sign in Eq. 9 depends on the moving direction, so S^{-1} in Alg. 1 takes sharpness and direction as inputs. Our adjustment process resolves defocus uncertainty and facilitates convergence within the DoF. This ensures a stable autofocus process across varying defocus conditions.

4. Evaluation

4.1. Experiment Setup

Experiments were conducted on an Automatic Optical Inspection (AOI) machine equipped with a COOILENS WWK15-110-111 lens and a DaHeng MER2-2000-6GM-P camera. This setup provided a depth-of-field (DoF) of ± 0.14 mm and a measurement range of ± 5 mm, corresponding to 35 times the DoF. Ground truth data was obtained from high-precision position sensors within the motion modules, offering a measurement precision of $1 \mu\text{m}$, ensuring highly accurate reference values and eliminating the need for synthetic defocus features.

The model was trained and tested on anodized objects, metal surfaces, printed paper, and printed circuit boards (PCB), which are common in industrial inspection, covering a diverse range of material textures to evaluate generalization performance. For both training and testing, focal stacks were captured across the ± 5 mm range with a step size of 0.1 mm, ensuring coverage of varying defocus levels. This process resulted in 480 focal stacks for training and 800 for testing, each containing 101 images with the focal plane set at 0 mm. The images were resized to 512×512 pixels, maintaining sufficient detail while optimizing computational efficiency. The model was trained for 50 epochs using the Adam optimizer with a learning rate of $2e-5$, implemented on an RTX 4090 GPU.

4.2. Comparison with Conventional Methods

This section compares our autofocus (AF) method with conventional approaches. As mentioned in Sec. 1, conventional AF methods are sensitive to texture variations. Fig. 4a shows sharpness curves generated using the Normalized-Variance algorithm (which achieved the best overall score in a previous study [33]), highlighting this sensitivity. Even within the same sample domain, peak values and curve shapes vary significantly, making conventional methods reliant on peak-searching. This variability is the primary cause of their low efficiency. In contrast, our method computes sharpness using Eq. 5 with a deep learning model, ensuring greater robustness across different textures. As shown in Fig. 6a, the peak values and overall curve shapes remain consistent across sample domains. This stability allows for more effective application of Alg. 1 in focus tuning, improving efficiency.

In this experiment, we adopt the coarse-to-fine peak search approach for conventional approaches due to its robustness [40]. In the search method, the coarse step size should be smaller than the *Range*, which is defined by the two local minima surrounding the sharpness peak [33]. To balance the success rate and sampling time, we fine-tuned this step size for each algorithm, denoted as *size* in Tab. 1, while setting the fine step size to match the DoF.

Table 1. Comparison with Conventional Autofocus

Method(Conventional)	Size(mm)	Coarse	Fine	Total	In-focus
Absolute Grad[33]	0.9	6	4	10	0.748
Variance[33]	1.2	5	5	10	0.771
Nor-Variance[33]	1.3	4	5	9	0.892
Auto Correlation[33]	0.8	7	3	10	0.774
STD Correlation[33]	1.3	4	5	9	0.861
Entropy[33]	2.6	2	10	12	0.855
Fisher Entropy[23]	0.8	7	3	10	0.898
Squared Gradient[33]	0.8	7	3	10	0.902
Modified DCT[16]	0.8	7	3	10	0.93
Diagnose Laplacian[5]	0.9	6	4	10	0.701
Tenengrad Var[24]	0.9	6	4	10	0.724
Laplacian Var[24]	0.8	7	3	10	0.874
Brenner Grad[33]	0.8	7	3	10	0.901
Tenenbaum Grad[33]	0.9	6	4	10	0.738
Modified Laplacian[33]	0.8	7	3	10	0.858
Energy Laplacian[33]	0.8	7	3	10	0.861
Method(Proposed)	Direction	Coarse	Fine	Total	In-focus
Ours	o	*	*	3	0.912
Ours	x	*	*	5	0.962

o: Initial Direction Correct, x: Initial Direction Wrong, *: Not Applicable

The o and x in Tab. 1 denote the initial adjustment direction of the proposed method. The x means wrong initial direction and usually requires one additional step to adjust it, as Fig. 6a shows. The focal plane was manually set to 0 mm by an expert. Given the sampling interval of 0.1 mm (significantly smaller than the DoF of 0.28 mm) and minor sample flatness variations, we defined the in-focus region as (± 0.21 mm) ($1.5 \times$ DoF) around 0 mm. To evaluate AF performance under severe defocus conditions, we randomly initialized starting positions within -5 mm to -4 mm or 5 mm to 4 mm across all methods, corresponding to a defocus range of $28 \times$ to $35 \times$ DoF. All methods were evaluated on a comprehensive test set to ensure a fair comparison.

Results are presented in Tab. 1. The proposed method consistently outperformed conventional approaches, even when the initial adjustment direction was incorrect. Since only a few conventional methods achieved a success rate above 90%, we selected the first step at which the success rate exceeded 90% as the basis for comparing the proposed and conventional methods. When the initial adjustment direction is correct (o), our approach surpasses 90% in-focus accuracy by step 3. Even when starting with the wrong initial direction (x), our method outperforms conventional approaches, reaching over 95% in-focus accuracy by step 5. These results highlight the robustness of our approach in handling severe defocus conditions, regardless of the initial adjustment direction.

4.3. Comparison with DL-based Methods

The main limitation of existing DL-based AF methods is defocus uncertainty. As shown in Fig. 5, defocused images from the left and right sides of the focal plane appear visually identical, making direct distance estimation prone to errors. As Fig. 4b illustrates, DE models risk predicting the distance in the wrong direction, leading to AF failure.

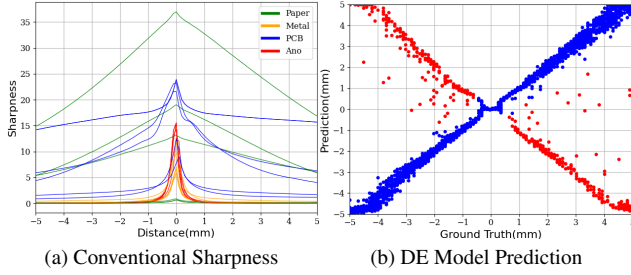


Figure 4. (a) The sharpness curves from the conventional method are sensitive to the texture variations. (b) Prediction results from a DE model. Red is wrong predictions due to defocus uncertainty.

The red data points in Fig. 4b indicate predictions on the incorrect side of the focal plane. In contrast, the proposed method determines direction through sharpness change, as described in Sec. 3.4. This ensures that the risk of incorrect direction prediction is limited to the initial step.

Initial training of compared models on our dataset resulted in poor performance, with an in-focus rate below 30% in 5 adjustments. To ensure a fair comparison, we trained and tested the compared models within the same domain for each comparing items (for example, both trained and tested on the metal dataset), while our method was trained on the comprehensive dataset (anodized, metal, printed-paper, PCB), demonstrating better generalization. We evaluated two models from Depth-from-Defocus studies [22][19], two from DL-based AF studies [17][10], and one vision transformer-based depth estimation model [27]. Each method was tested with five adjustment iterations, reporting the highest in-focus rate achieved. Tab. 2 shows that our method consistently outperforms the compared models under the small DoF setting at the starting position from severe defocus range.

Table 2. Comparison with DL-based Methods

Methods	Anodized	Metal	Paper	PCB
Nazir et al.[22]	0.846	0.847	0.759	0.859
Liao et al.[17]	0.19	0.504	0.628	0.565
Lu et al.[19]	0.58	0.33	0.23	0.423
Herrmann et al.[10]	0.455	0.83	0.788	0.835
Ranftl et al.[27]	0.688	0.305	0.38	0.824
Ours(o)	0.998	0.882	0.947	0.965
Ours(x)	0.994	0.867	0.912	0.919

o: Initial Direction Correct, x: Initial Direction Wrong

4.4. Evaluation of Domain Generalization

This section evaluates the robustness of the proposed autofocus method across different material textures and illumination conditions. The results demonstrate the method’s ability to maintain high in-focus accuracy across diverse real-world industrial inspection scenarios from severe defocus starting positions.

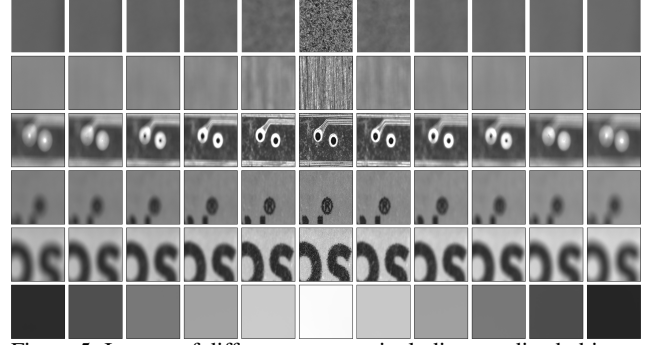


Figure 5. Images of different textures, including anodized objects, metal, PCB, and printed paper. From left to right: defocused images from the near side to the lens to the far side.

Different Textures: The proposed method was evaluated across different textures. Tab. 3 presents the in-focus rate at each step, and Fig. 5 shows sample images for each texture. The first row is anodized samples containing rich high-frequency textures and uniform surface reflectiveness. The second row is metal samples. Which also have high-frequency texture across the surface. However, its high reflectivity makes it easy to see saturated pixels in images. The third row is PCB samples, which only have a small area containing high reflectivity features. The fourth and fifth rows are printed paper with black content and a white background. The final row is the average sharpness output from the U2-Net model. All the samples have different features and are common in industrial inspection tasks.

Table 3. In-focus Rate of Different Textures

Steps	1	2	3	4	5	6	7
Ano(o)	0.598	0.835	0.996	0.998	1.000	1.000	1.000
Ano(x)	0.000	0.209	0.986	0.994	1.000	1.000	0.998
Metal(o)	0.218	0.841	0.882	0.945	0.986	0.972	0.972
Metal(x)	0.000	0.107	0.867	0.725	0.931	0.960	0.950
Paper(o)	0.421	0.439	0.719	0.947	0.833	0.974	0.991
Paper(x)	0.000	0.430	0.456	0.623	0.912	0.825	0.904
PCB(o)	0.663	0.895	0.965	0.988	0.988	0.977	0.988
PCB(x)	0.000	0.047	0.919	0.907	0.977	0.988	0.988

o: Initial Direction Correct, x: Initial Direction Wrong

Our method performs best on anodized and PCB samples due to their rich textures, achieving over 90% in-focus accuracy by step 3. For metal samples, performance drops slightly compared to anodized and PCB samples, reaching 85% in-focus accuracy by step 3, likely due to the high reflectivity of metals. Printed paper samples pose the most significant challenge, achieving 71.9% and 45.6% accuracy under two initial conditions at step 3. However, both cases exceed 90% in-focus accuracy in steps 4 and 5. The high contrast between the white background and black content may introduce difficulties in sharpness prediction, affecting autofocus performance. Despite these variations, our method consistently converges to high in-focus accuracy across different textures.

Table 4. In-focus Rate of Metal Under Different Illuminations

Steps	1	2	3	4	5	6	7
Dark(o)	0.183	0.866	0.923	0.986	1	0.993	0.993
Dark(x)	0.000	0.077	0.951	0.761	0.993	1	1
Normal(o)	0.366	0.824	0.958	0.993	1	0.993	1
Normal(x)	0.000	0.127	0.944	0.887	0.986	1	1
Bright(o)	0.141	0.761	0.754	0.852	0.965	0.908	0.915
Bright(x)	0.000	0.085	0.711	0.542	0.831	0.93	0.859

o: Initial Direction Correct, x: Initial Direction Wrong

Different Illuminations: To evaluate the robustness of our method under varying illumination conditions, we conducted experiments on metal samples due to their high reflectivity. We collected three sets of samples under bright (average intensity 168), normal (128), and dark (88) conditions. The predicted sharpness values for each condition are shown in Fig. 6b. Under bright conditions, some predictions appear abnormal, likely due to image saturation.

Since our method derives the adjustment distance from the inverse function of sharpness (Eq. 9), variations in sharpness predictions directly impact autofocus performance. As shown in Tab. 4, the in-focus rate under bright conditions is the lowest, reaching only 70% by step 3, whereas both normal and dark conditions exceed 90% at the same step. We assume that under bright illumination, saturation effects disrupt sharpness prediction, leading to occasional AF failures, as shown in Fig. 6b. However, by steps 5 and 6, the in-focus rate still reaches 90% across all conditions, demonstrating the robustness of our method against illumination variations and reflective surfaces. This finding demonstrates the method’s capability to address common challenges in industrial environments.

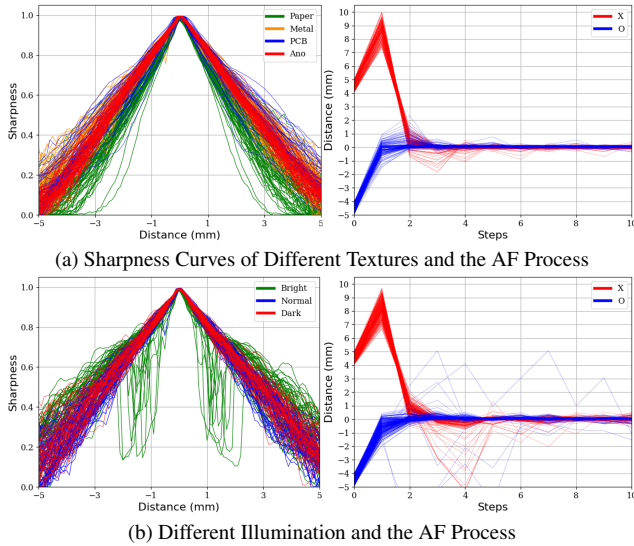


Figure 6. (a) Consistent sharpness across textures results in a stable AF process. (b) Illumination variations in metal samples, particularly under bright conditions, lead to slightly less stable sharpness and AF performance.

4.5. AF in All Range and Severe Defocus

In this section, we present the detailed results of the proposed AF method on the comprehensive test dataset under different starting positions. The *all range* in Tab. 5 corresponds to autofocus starting positions from -5 mm to 5 mm. When the initial direction is correct, the in-focus rate reaches 64.3% at the first step, 90% at step 3, and exceeds 95% at step 5. When the initial direction is incorrect, performance is significantly worse in the first two steps but eventually reaches 95% at step 5.

For the severe defocus range (28× to 35× DoF), corresponding to initial positions between -5 to -4 mm and 4 to 5 mm, our method achieves 90% and 85% in-focus rates at step 3, respectively, and eventually exceeds 95%. This performance closely matches that of the *all range* case, demonstrating that our method remains equally effective even from highly defocused starting points. Overall, the proposed method maintains high autofocus accuracy even under severe defocus conditions, with no significant drop in performance compared to the all-range case. Achieving 95% in-focus accuracy at step 5 in both cases demonstrates its suitability for small DoF scenarios such as industrial inspection and microscopy.

Table 5. In-focus Rate of All and Severe Defocus (SD) Range

Steps	1	2	3	4	5	6	7
All(o)	0.643	0.807	0.939	0.978	0.967	0.981	0.985
All(x)	0.027	0.521	0.860	0.896	0.970	0.949	0.964
SD(o)	0.428	0.781	0.912	0.969	0.975	0.976	0.982
SD(x)	0.000	0.169	0.872	0.843	0.962	0.965	0.961

o: Initial Direction Correct, x: Initial Direction Wrong,

4.6. Evaluation of Sharpness Estimation

Optical model-based sharpness estimation is crucial for autofocus, as focus adjustment relies on the inverse function of sharpness (Eq. 9). This section examines the effectiveness of deep learning-based sharpness regression, comparing different model architectures and analyzing how texture and illumination conditions influence sharpness prediction and autofocus performance.

We evaluate several deep learning architectures for sharpness regression, including VIT [27], U-Net [28], Res-Net [36], Dense-Unet, and U2-Net [26]. The CNN-based models follow a five-layer encoder-decoder structure, while VIT utilizes a self-attention mechanism. Experiments on 800 focal stacks show that U2-Net achieved the best overall performance, as Tab. 6 shown. The details of the evaluation metrics can be found in Caden et al. [2].

Fig. 6a illustrates the relation between predicted sharpness and the distance across different textures. While deviations from the ground truth increase with severe defocus, sharpness values consistently converge near the focal plane regardless of texture. This consistency confirms that

Table 6. Sharpness Prediction Performance Across Different Models

	MAE	MSE	RMSE	LogMSE	LogRMSE	AbsRel	SqrRel	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
U-Net	0.0582	0.008	0.0585	0.0009	0.0188	0.0429	0.0061	0.732	0.8431	0.8927
Res-Unet	0.06	0.0081	0.0606	0.00089	0.0194	0.0435	0.0059	0.708	0.8283	0.8865
Dense-Unet	0.0607	0.0085	0.0608	0.00095	0.0194	0.0443	0.0063	0.7224	0.8356	0.8878
U2-Net	0.057	0.0076	0.0573	0.00083	0.0182	0.0416	0.0057	0.7322	0.8508	0.9028
VIT	0.0639	0.0089	0.0657	0.001	0.0208	0.0486	0.007	0.7194	0.8451	0.904

Table 7. Sharpness Evaluation Across Different Textures and Illumination Using U2-Net

Different Textures										
	MAE	MSE	RMSE	LogMSE	LogRMSE	AbsRel	SqrRel	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
Ano	0.0374	0.0026	0.0375	0.0003	0.0119	0.0269	0.0019	0.8052	0.9106	0.9511
Metal	0.0601	0.0083	0.0603	0.0009	0.0191	0.0456	0.0065	0.7250	0.8499	0.9053
Paper	0.1166	0.0227	0.1169	0.0025	0.0376	0.0820	0.0162	0.4878	0.6476	0.7348
PCB	0.0348	0.0033	0.0359	0.0003	0.0112	0.0254	0.0025	0.8399	0.9186	0.9497
Different Illumination of Metal Samples										
Bright	0.0839	0.0147	0.0843	0.0015	0.0263	0.0629	0.0112	0.6478	0.7966	0.8698
Normal	0.0418	0.0038	0.0420	0.0004	0.0135	0.0317	0.0030	0.7874	0.8918	0.9350
Dark	0.0540	0.0062	0.0541	0.0007	0.0174	0.0418	0.0052	0.7420	0.8629	0.9121

the sharpness trends remain stable across different surfaces, making depth estimation more predictable. However, inaccurate sharpness predictions lead to incorrect distance estimations, affecting autofocus performance.

Texture-dependent sharpness estimation performance, detailed in Tab. 7, directly impacts autofocus success rates. Anodized and PCB samples exhibit the lowest estimation errors with higher in-focus rates and fewer adjustment steps. In contrast, metal samples show moderately higher errors due to reflective surfaces, while printed paper samples present the most significant challenges as high-contrast regions disrupt sharpness consistency. As shown in Tab. 3, these less reliable estimates require additional refinements to reach acceptable focus levels; overall, however, the proposed approach achieves robust autofocus performance across all textures within a few steps.

Illumination conditions, especially in highly reflective samples, also affect sharpness estimation. In bright settings, saturation effects degrade model performance. As shown in Tab. 4 in the illumination section, metal samples under bright lighting exhibit the highest prediction errors, aligning with our assumption in Sec. 4.4 that overexposure reduces sharpness reliability.

Overall, the accuracy of sharpness estimation is directly linked to autofocus performance. The results confirm that U2-Net provides the most reliable sharpness predictions across different textures and lighting conditions, improving autofocus efficiency. Poor sharpness regression, particularly in highly reflective or high-contrast surfaces, negatively impacts autofocus performance by increasing the required adjustment steps. These findings emphasize the importance of robust sharpness estimation in ensuring precise and efficient autofocus in small DoF imaging systems.

5. Conclusion

In this work, we proposed a novel autofocus (AF) method to address the challenges associated with small DoF scenarios and severe defocus conditions. By integrating an optical model-based sharpness indicator with a deep learning framework, our approach enables distance estimation even from severely defocused images through sharpness-distance mapping, achieving autofocus without dense sampling and avoiding defocus uncertainty across a $35\times$ DoF range.

Extensive evaluations demonstrate that our method outperforms conventional image-based and existing DL-based AF techniques in accuracy, efficiency, and adaptability. On a comprehensive test dataset containing anodized objects, metal surfaces, printed paper, and PCB, our method achieves over 95% in-focus accuracy within five iterations, effectively handling highly defocused starting positions while maintaining stability across diverse material surfaces common in industrial inspection. This highlights its suitability for small DoF scenarios such as industrial inspection and microscopy. By overcoming key limitations of existing AF methods, our approach enhances autofocus reliability in applications requiring precise focus control. Future work will explore its extension to dynamic environments and multi-plane focusing systems to further improve its capabilities.

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